Strategic Serial Processing:

Bottleneck Models of Psychological Refractory Period Paradigms

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It is well established that not only do people struggle to focus on two tasks at once, but also perform poorly in making two consecutive judgements. These limits of human processing capacity have been theorised to occur due to reliance on a decision-making process which operates in serial (Welford, 1952). This so-called 'bottleneck' in cognitive processing accounts for the fact that people perform best on a single task at a single time. The following paper sheds light on this serial decision-making mechanism by reviewing literature on the psychological refractory period (PRP), providing qualifications to the classic bottleneck model which posits that the PRP effect is caused by serial processing at the response selection stage. Instead, I make the case that cognitive systems adapt to task (stimulus and response) demands and strategically divide attentional resources that contribute to PRP effects.

Psychological Refractory Period Paradigm

The psychological refractory period (PRP) was first observed in 1931 (Telford) and has been explored by psychologists for over seven decades. The PRP is a phenomenon in which a delayed response on a second task is observed when it is immediately followed by a prior task that demands cognitive resources. In other words, when subjects participate in a task (T_1) requiring a response (R_1) which is followed by another task (T_2) that also requires a response (R_2), performance measured in reaction time (RT) is usually more degraded for T_2 when the time between the onset of the first stimulus and the second stimulus is shorter (Pashler, 1984). The duration of this stimulus-onset asynchrony (SOA) predicts the magnitude of the PRP effect. Crucially, the stimuli for T_2 present themselves before R_1 is complete at short SOA intervals – hence why the PRP paradigm is also known as the 'overlapping tasks paradigm' (Navon &

Miller, 2002).

In a PRP paradigm, participants are given a set of tasks that require them to differentially respond to certain types of stimuli, such as pressing a button if a number is odd, another button if a number is even, a third button if a letter is a consonant and a fourth if it is a vowel (Lien et al., 2003; see Figure 1a). To test the PRP effect, participants are first presented with an odd number, rapidly followed by a consonant. The RT_1 for response R_1 to the odd number remains virtually constant across SOA conditions; however, the response R_2 to the consonant is most delayed when the SOA between the stimuli is short (see Figure 1b). That is, when the SOA between T_1 and T_2 is longer, the time delay for R_2 is reduced. Similar results are seen across a wealth of PRP studies and their variations, including studies when the task and response are coded for motor stimuli (Hibberd et al., 2010), when the task load is shared across two people (Liepelt & Prinz, 2011), and when individual differences like personality or intelligence are controlled for (Indermühle et al., 2011).

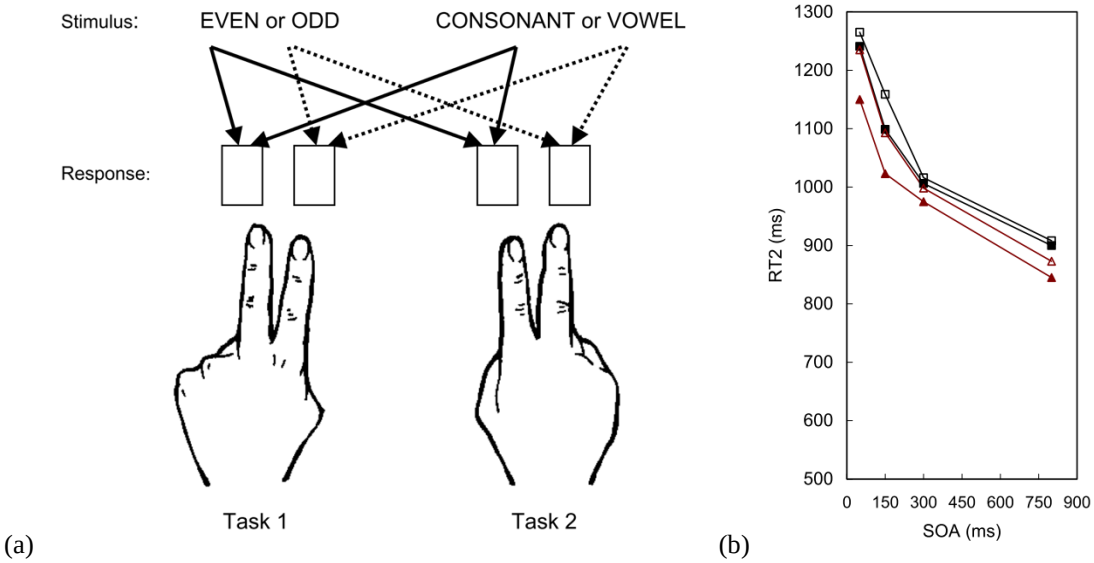


Figure 1. (a) The stimulus and response pairings for a PRP task and **(b)** RT_2 for response to the second stimulus R_2 as a function of SOA between T_1 and T_2 as in Lien et al. (2003).

Serial Processing and Bottleneck Models

A seminal account for this robust phenomenon comes from Welford's (1952) bottleneck model, which he later referred to as a *single channel hypothesis* (Welford, 1967). He proposed that this lag in RT for R_2 implies that there is some kind of processing to be completed for T_1 before the system can move onto T_2 (see Figure 2). This mechanism can only handle a single task at once, hence more time is taken for R_2 at short SOAs because part of this time is initially contributed to making R_1; whereas in long SOA conditions, the time required for R_1 is spent in the SOA itself and hence reduces the use of time required for R_1 after T_2 is presented, enabling R_2 to be made quicker. A simple analogy for this model is a queue with a single cashier (bottleneck): one customer has to be checked out (R_1) before the next (R_2) can be addressed, and if the second customer joins the queue immediately after the first one (short SOA) the waiting time (delay in RT_2) will be longer than if they joined the queue later (long SOA, shorter RT_2). In other words, the longer RT_2 is simply due to a *postponement* of response processing due a hold-up in the bottleneck caused by the processing of T_1.

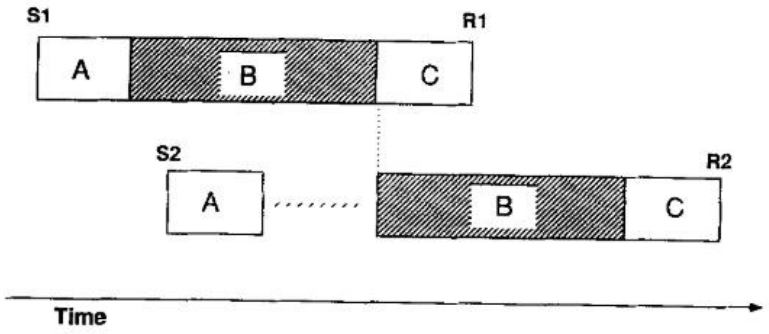


Figure 2. A visualization of the bottleneck model of the PRP effect as in Pashler (1994). The shaded portion represents the bottleneck of serial processes that cannot overlap.

The question arises, then, about which part of the response process is causing this postponement. Pashler (1994) suggested that making a response to T_1 and T_2 involves three

stages: (1) perceiving the stimulus, (2) deciding on the correct response and (3) performing or executing the response. Since a plethora of visual search task studies have shown that multiple stimuli can be perceived at one time (Chen, 2012; Egeth & Yantis, 1997), it is obvious that the bottleneck is not at the initial perception stage. It is more difficult to distinguish whether the bottleneck is at the response-selection or response-execution stage. These have been empirically tested by manipulating the complexity of T_1 or complexity of R_1 respectively – the logic being that a more complex task will prolong response selection stage by making it more difficult to arrive at an answer, but a more complex response will prolong response execution by making it more difficult to perform the response. The prolonging of the stage that causes additive effects in R_2 can be concluded to be the bottleneck stage, while the other stage that produces 'underadditive' effects is not.

Much research has produced such additive effects when the response-selection stage is prolonged, leading to the *response-selection bottleneck* (RSB) model remaining the dominant theory that accounted for the PRP during much of the 1990s (McCann & Johnston, 1992; Miller & Durst, 2015; Pashler & Johnston, 1989). Another method that has been used to test whether the response-selection bottleneck exists presents an unexpected stimulus along with the T_2 stimulus; showing that delay of RT_2 is resistant to involuntary triggers (Maslovat et al., 2013). However, these RSB studies were restricted to auditory and visual stimuli and simple motor or vocal responses – manipulating the type of stimuli and response did not produce similarly clear evidence for the RSB model. Ayora et al. (2011) found underadditive effects when T_2 required them to respond to a picture, while ignoring a word simultaneously presented on the screen. Thus, semantic interference is not adequately explained by the RSB model, since a picture-word compound at T_2 should increase RT_2. PRP effects were also found to be more pronounced with

certain paired input-output modalities, like auditory-manual tasks, than others (Hazeltine et al., 2006; Huestegge & Hazeltine, 2011), which may explain the compelling findings of the 1990s studies that mainly used auditory input and motor output.

When there is a congruence between the response for T_1 and T_2, the additive effects of task difficulty are also decreased – that is, when there is a spatial similarity in the motor response for both the tasks, there is less of a lag before R_2 (Schubert et al., 2008). This finding suggests that the process of response execution contributes to PRP, because incongruent responses increase and congruent responses reduce RT_2 (see Figure 3), which would not be the case if response activation occurred in parallel. Bratzke (2008) further manipulated type of motor responses demanded by requiring T_1 to perform a ballistic movement and T_2 a simple key-press. This revealed a 'propagation effect' wherein RT_2 increased with time taken to execute the ballistic R_1 movement, once again suggesting that response execution is serial – the time taken to complete the R_1 movement caused a lag in RT_2.

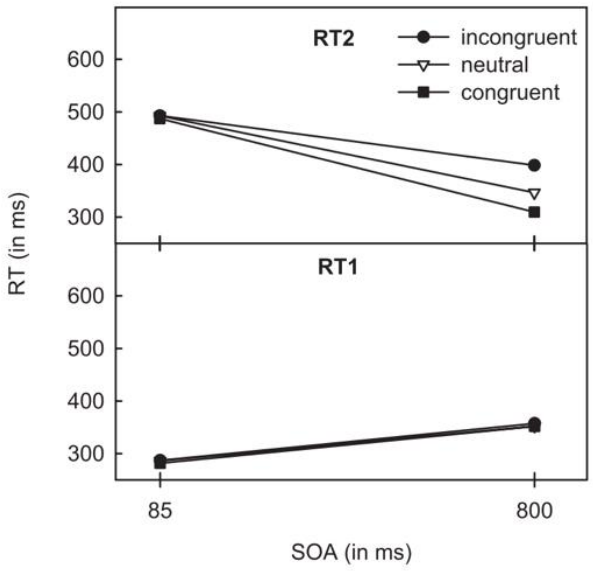


Figure 3. Mean RT on the PRP paradigm as a function of short and long SOA as in Schubert et al. (2008).

It is important to note that the evidence from the aforementioned studies does not

disprove the RSB model, but merely suggests that there may be interactive effects between the tasks (known in the literature as 'crosstalk') that externally contribute to the additive effects in PRP, or that two bottlenecks may exist at response-selection and response-execution whose limitations vary depending on the type of and relationship between stimulus and response as well as task demands (Fischer & Plessow, 2015; Navon & Miller, 2002). The claims that serial processing occurs at both response-selection and response-execution stages has been backed up by similar neuroimaging results at both stages (Sommer et al., 2001; Yildiz et al., 2013).

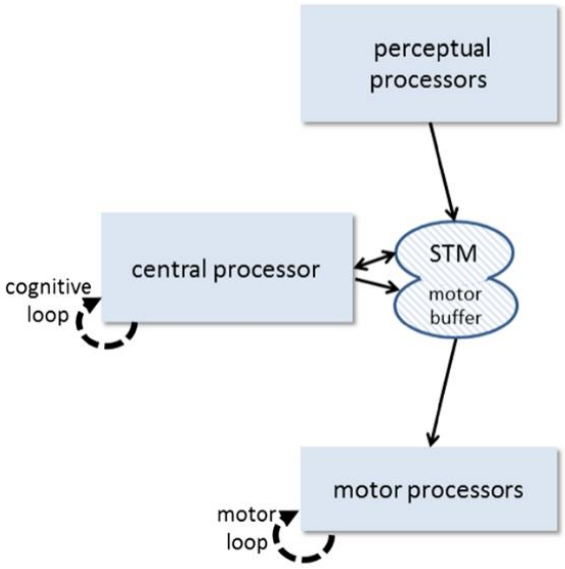


Figure 4. The Cognitive framework for Sequential Motor Behaviour (C-SMB) by Verwey et al. (2014).

However, more studies have begun attributing the bottleneck entirely to the response-execution stage (Bratzke et al., 2009; Klapp et al., 2019), leading scholars to posit various theories of sensorimotor perception and execution. Verwey et al. (2014) offered a cognitive framework for sequential execution strategies that attempts to explain serial processing in motor response-execution for the PRP paradigm. This framework includes separate representational codes and processors for visual, auditory and motor input that are handled by a central processor

through the short-term memory (STM) store (see Figure 4). Feng et al. (2014) suggested that cognitive systems engage in 'multiplexing' through which different processors use the same sensory representations to carry out operations, which may restrict parallel response-execution.

Strategic Influences on PRP

The influence of task demands, modality and response type on the PRP paradigm implies that the bottleneck mechanism is more versatile than previously thought. This versatility means that the bottleneck is not an inherent capacity limit built into our cognitive systems, but a mechanism that can be strategically employed to improve performance on tasks. Researchers have theorised that the bottleneck mechanism is a strategic diversion of attention, providing a multitude of factors that provide evidence for the claim that cognitive attentional resources can be employed to reduce the PRP effects, which have been listed below.

It has been found that participants are capable of learning how to reduce the RT_2 disadvantage in the PRP paradigm. Garner at al. (2014) showed that even a single practice session can improve performance on the PRP. The more practice sessions a participant goes through, the more the task becomes automatic to them, and fewer the attentional costs they incur: 6 practice sessions reduced the duration of the RT_2 period by 170 milliseconds (Maquestiaux et al., 2008), whereas 36 practice session offered a reduction of 313 milliseconds, with the PRP effects virtually disappearing for one participant (Van Selst et al., 1999). Interestingly, training that improved response-selection also improved perceptual consolidation, suggesting that there is an interactive effect between the perceptual encoding and subsequent response-selection as well (Garner et al., 2014).

Accuracy and RTs also improve when participants are monetarily motivated to win – participants who were given money and told they would lose some of it for every wrong answer

on a trial had faster RTs than people who were not given any trial-dependent incentives (Yildiz et al., 2013). Even without explicit instructions, Fischer and Dreisbach (2015) found that people learned to predict the SOA while responding to stimuli in the PRP paradigm when specific stimuli on T_1 were coded to precede a long SOA and some were coded to predict a short SOA. Delay in RT_2 was significantly reduced when a short SOA was anticipated (Fischer & Dreisbach, 2015). All this evidence suggests that people are capable of strategically diverting attentional resources to eliminate the cognitive bottleneck when provided with sufficient training or motivation.

Individual differences in cognitive capacity also predict PRP effects in people, presumably due to people with higher cognitive capacities being able to reduce PRP effects by more strategically deploying attentional resources to overcome the bottleneck. Lower PRP is associated with higher scores on processing speed, inhibition, flexibility and working memory tests (Laguë-Beauvais et al., 2013). Practice effects may also be partially attributed to a general improvement in executive control skills, which may be responsible for the better allocation of cognitive attentional resources that help in overcoming the bottleneck and decrease the PRP effects (Strobach & Schubert, 2017). Thus, the bottleneck mechanism in PRP paradigms is not an inherent capacity-limit, but can be overcome with strategic deployment of cognitive resources.

Conclusion

The psychological refractory period paradigm is one of the oldest and most robust paradigms in cognitive psychology. The PRP generates a delayed response time to a stimulus that immediately follows a prior stimulus that requires a different task response by the participant. Classical theories suggest that this delay is due to a ‘bottleneck’ mechanism that

must allow the response for the first stimulus to be correctly decided before it can move on to deciding a response for the second stimulus. However, further research has shown that this bottleneck may not be exclusive to response-selection, but may also extend to the behavioural response-execution stage. This latter theory has been widely supported by evidence that has found that PRP effects vary as a function of manipulation of the type and nature of motor responses required between the first and second task. This, paired with other evidence for strong influences of modality and demands of the task on PRP, imply that the bottleneck is not an inherent cognitive capacity-limiter, but can be adjusted to meet task demands. This adjustment is based on the amount of attention a participant deploys to monitor their performance on the PRP paradigm: people are capable of reducing PRP effects when sufficiently trained or motivated to divert attentional resources.

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